Extra Course B: Receiver Measurement and Performance

VNAs, S-parameters, noise figure, MDS, dynamic range, and the receiver vocabulary that makes ham radio measurement intelligible.

Groups E4B + E4C · 25 pool questions · 30-45 minute presentation · Foundation for Bite #13 · Practice exams at eham.net/exams

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Part 1: Measurement Instruments (E4B)

Frequency counter accuracy
A frequency counter's accuracy is set by the accuracy of its **time base**. (E4B01) The counter is just counting cycles in a known time window; if the window is wrong, every reading is wrong by the same fraction.

Voltmeter sensitivity ("ohms per volt")
An analog VOM's specification "ohms per volt" tells you its DC input impedance per range. **Full-scale reading × ohms-per-volt = input impedance of the voltmeter.** (E4B02) A 20,000 ohms/volt meter on the 10 V scale is 200 kΩ input – high enough to not load most circuits much.

Vector Network Analyzer (VNA) basics
A VNA is the standard tool for measuring two-port RF networks (filters, amplifiers, antennas). It generates a known RF signal at the input port, measures both reflected and transmitted waves at both ports, and computes S-parameters across frequency.

S-param	Meaning	Q
S11	Input port return loss / reflection coefficient (equivalent to VSWR at port 1). (E4B04)	E4B04
S21	**Forward gain** (signal from port 1 to port 2). (E4B03)	E4B03
S12	Reverse isolation (port 2 back to port 1).	—
S22	Output port return loss.	—

S-parameter subscripts = **the port or ports at which measurements are made**. (E4B07) S21 means "output at port 2, input at port 1."

VNA calibration uses three standard test loads: **short circuit, open circuit, and 50 ohms**. (E4B05) These three references let the instrument back out cable and connector effects so it measures only the device under test.

A two-port VNA can measure **filter frequency response** (E4B09), and more broadly: **input impedance, output impedance, reflection coefficient – all of the above**. (E4B11)

Power-meter math: forward minus reflected

Power absorbed by the load = Forward power − Reflected power.
Example: 100 W forward, 25 W reflected ⇒ load absorbs 100 − 25 = **75 W**. (E4B06)

Measuring Q of a series-tuned circuit
The Q of a tuned circuit can be determined from **the bandwidth of the circuit's frequency response**. (E4B08) Specifically, $Q = f_0 / BW$ (the inverse of the formula used in EC-A).

Two-tone intermodulation distortion measurement
Standard method for an SSB transmitter: **modulate the transmitter with two AF (audio) signals at non-harmonically related frequencies, and look at the RF output with a spectrum analyzer**. (E4B10) The IMD products show up as extra spectral lines on either side of the wanted output.

Part 2: Receiver Performance (E4C)

Noise floor and noise figure

Concept	Definition / value	Q
Noise figure (NF)	The ratio in dB of the noise generated by the receiver to the theoretical minimum noise. Lower NF = quieter receiver.	E4C04
−174 dBm noise floor	The **theoretical noise in a 1 Hz bandwidth** at the input of a perfect receiver at room temperature. The reference number for noise-floor calculations.	E4C05
Bandwidth increase 50 Hz → 1,000 Hz	Noise floor rises by **13 dB**. (1000/50 = 20×; 10·log(20) = 13 dB.)	E4C06
MDS (Minimum Discernible Signal)	The smallest signal the receiver can pull out of its noise floor. (E4C07)	E4C07

SDR overload
A direct-sampling SDR is overloaded when the input signal exceeds the **reference voltage of the ADC**. (E4C08) Past that point the ADC clips and the signal is destroyed.

Phase noise and reciprocal mixing

Concept	What it does	Q
Phase noise (SDR master clock)	Can **combine with strong nearby signals to generate interference**. (E4C01)	E4C01
Reciprocal mixing	**Local oscillator phase noise mixes with adjacent strong signals to create interference to desired signals.** (E4C13)	E4C13

Selectivity and adjacent-signal handling

Feature	What it does	Q
Front-end filter / preselector	Eliminates interference from strong **out-of-band** signals before they reach the mixer.	E4C02
High IF in a superhet receiver	Makes it **easier for front-end circuitry to eliminate image responses**. (Higher IF pushes the image further from the wanted signal.)	E4C09
Variety of bandwidths	Lets you match the receive bandwidth to the modulation, **maximizing SNR and minimizing interference**.	E4C10
Narrow roofing filter	Improves blocking dynamic range by attenuating strong signals near the receive frequency before they hit the IF amplifier and AGC.	E4C12
IF Shift control	**Reduces interference from stations transmitting on adjacent frequencies** by shifting the IF passband relative to the wanted signal.	E4C14

Capture effect
In an **FM receiver**, when two signals are on the same frequency, the stronger one **suppresses** the weaker. The phenomenon is called **capture effect**. (E4C03)

Attenuators at low HF
Why does adding receiver input attenuation reduce overload on the lower HF bands without hurting SNR?
Atmospheric noise is generally greater than internally generated noise even after attenuation. (E4C11) The external noise dominates, so attenuating both signal and noise together (and the receiver's quiet internal noise) keeps the SNR almost unchanged but tames the dynamic-range problem.

Mental Cheat Sheet

- Frequency counter accuracy = time base accuracy.
- Voltmeter input Z = full-scale reading × ohms-per-volt rating.
- VNA calibration: short, open, 50 ohms.
- S-parameters: S11 = return loss; S21 = forward gain. Subscripts = ports.
- Two-port VNA measures filter response, input/output Z, reflection coefficient (all of the above).
- Power absorbed = Forward − Reflected.
- Q from response = f_0 / BW.
- IMD test for SSB: two non-harmonic AF tones, look at RF on a spectrum analyzer.
- Noise figure = receiver noise / theoretical minimum, in dB.
- −174 dBm = theoretical noise in 1 Hz at room temperature.
- BW 50 → 1000 Hz raises noise floor by 13 dB.
- MDS = minimum discernible signal.
- SDR overloads when input exceeds the ADC reference voltage.
- Phase noise causes interference from strong nearby signals; reciprocal mixing is the LO version.
- Preselector blocks out-of-band; roofing filter blocks near-band; IF shift dodges an adjacent station.
- Capture effect: FM, stronger signal wins on same frequency.
- HF attenuator works because atmospheric noise dominates.

Common Traps

- S21 = forward gain (not S12, which is reverse isolation).
- VNA calibration is three loads: **short, open, 50 ohms** (not "50/75/90" or three lengths of coax).
- Power absorbed = Fwd − Refl (not Fwd + Refl or Fwd alone).
- IMD test uses **two AF tones, non-harmonically related**, with a spectrum analyzer at the RF output. The trap answers use harmonic tones, RF tones at the input, or a wattmeter.
- −174 dBm is per 1 Hz of bandwidth at room temperature.
- Roofing filter improves blocking dynamic range, not sensitivity.
- HF input attenuation works because atmospheric noise dominates, not because the attenuator filters interference.
- Capture effect is an FM phenomenon, not SSB or AM.

Takeaway: A VNA puts known RF into your device under test and measures the S-parameters back, calibrated against short/open/50 Ω references. Power absorbed = forward minus reflected. On the receiver side, noise figure and −174 dBm/Hz set the floor; bandwidth, MDS, dynamic range, preselectors, roofing filters, and IF shift give you the tools to dig real signals out of the chaos.

Extra Course B Question Bank (E4B + E4C)

E4B01. Which of the following factors most affects the accuracy of a frequency counter?
A. Input attenuator accuracy.
B. Time base accuracy.
C. Decade divider accuracy.
D. Temperature coefficient of the logic.

E4B02. What is the significance of voltmeter sensitivity expressed in ohms per volt?
A. The full scale reading of the voltmeter multiplied by its ohms per volt rating is the input impedance of the voltmeter.
B. The reading in volts multiplied by the ohms per volt rating will determine the power drawn by the device under test.
C. The reading in ohms divided by the ohms per volt rating will determine the voltage applied to the circuit.
D. The full scale reading in amps divided by ohms per volt rating will determine the size of shunt needed.

E4B03. Which S parameter is equivalent to forward gain?
A. S11.
B. S12.
C. S21.
D. S22.

E4B04. Which S parameter represents input port return loss or reflection coefficient (equivalent to VSWR)?
A. S11.
B. S12.
C. S21.
D. S22.

E4B05. What three test loads are used to calibrate an RF vector network analyzer?
A. 50 ohms, 75 ohms, and 90 ohms.
B. Short circuit, open circuit, and 50 ohms.
C. Short circuit, open circuit, and resonant circuit.
D. 50 ohms through 1/8 wavelength, 1/4 wavelength, and 1/2 wavelength of coaxial cable.

E4B06. How much power is being absorbed by the load when a directional power meter connected between a transmitter and a terminating load reads 100 watts forward power and 25 watts reflected power?
A. 100 watts.
B. 125 watts.
C. 112.5 watts.
D. 75 watts.

E4B07. What do the subscripts of S parameters represent?
A. The port or ports at which measurements are made.
B. The relative time between measurements.
C. Relative quality of the data.
D. Frequency order of the measurements.

E4B08. Which of the following can be used to determine the Q of a series-tuned circuit?
A. The ratio of inductive reactance to capacitive reactance.
B. The frequency shift.
C. The bandwidth of the circuit's frequency response.
D. The resonant frequency of the circuit.

E4B09. Which of the following can be measured by a two-port vector network analyzer?
A. Phase noise.
B. Filter frequency response.
C. Pulse rise time.
D. Forward power.

E4B10. Which of the following methods measures intermodulation distortion in an SSB transmitter?
A. Modulate the transmitter using two RF signals having non-harmonically related frequencies and observe the RF output with a spectrum analyzer.
B. Modulate the transmitter using two AF signals having non-harmonically related frequencies and observe the RF output with a spectrum analyzer.
C. Modulate the transmitter using two AF signals having harmonically related frequencies and observe the RF output with a peak reading wattmeter.
D. Modulate the transmitter using two RF signals having harmonically related frequencies and observe the RF output with a logic analyzer.

E4B11. Which of the following can be measured with a vector network analyzer?
A. Input impedance.
B. Output impedance.
C. Reflection coefficient.
D. All these choices are correct.

E4C01. What is an effect of excessive phase noise in an SDR receiver's master clock oscillator?
A. It limits the receiver's ability to receive strong signals.
B. It can affect the receiver's frequency calibration.
C. It decreases the receiver's third-order intercept point.
D. It can combine with strong signals on nearby frequencies to generate interference.

E4C02. Which of the following receiver circuits can be effective in eliminating interference from strong out-of-band signals?
A. A front-end filter or preselector.
B. A narrow IF filter.
C. A notch filter.
D. A properly adjusted product detector.

E4C03. What is the term for the suppression in an FM receiver of one signal by another stronger signal on the same frequency?
A. Desensitization.
B. Cross-modulation interference.
C. Capture effect.
D. Frequency discrimination.

E4C04. What is the noise figure of a receiver?
A. The ratio of atmospheric noise to phase noise.
B. The ratio of the noise bandwidth in hertz to the theoretical bandwidth of a resistive network.
C. The ratio in dB of the noise generated in the receiver to atmospheric noise.
D. The ratio in dB of the noise generated by the receiver to the theoretical minimum noise.

E4C05. What does a receiver noise floor of −174 dBm represent?
A. The receiver noise is 6 dB above the theoretical minimum.
B. The theoretical noise in a 1 Hz bandwidth at the input of a perfect receiver at room temperature.
C. The noise figure of a 1 Hz bandwidth receiver.
D. The receiver noise is 3 dB above theoretical minimum.

E4C06. How much does increasing a receiver's bandwidth from 50 Hz to 1,000 Hz increase the receiver's noise floor?
A. 3 dB.
B. 5 dB.
C. 10 dB.
D. 13 dB.

E4C07. What does the MDS of a receiver represent?
A. The meter display sensitivity.
B. The minimum discernible signal.
C. The modulation distortion specification.
D. The maximum detectable spectrum.

E4C08. An SDR receiver is overloaded when input signals exceed what level?
A. One-half of the maximum sample rate.
B. One-half of the maximum sampling buffer size.
C. The maximum count value of the analog-to-digital converter.
D. The reference voltage of the analog-to-digital converter.

E4C09. Which of the following choices is a good reason for selecting a high IF for a superheterodyne HF or VHF communications receiver?
A. Fewer components in the receiver.
B. Reduced drift.
C. Easier for front-end circuitry to eliminate image responses.
D. Improved receiver noise figure.

E4C10. What is an advantage of having a variety of receiver bandwidths from which to select?
A. The noise figure of the RF amplifier can be adjusted to match the modulation type, thus increasing receiver sensitivity.
B. Receiver power consumption can be reduced when wider bandwidth is not required.
C. Receive bandwidth can be set to match the modulation bandwidth, maximizing signal-to-noise ratio and minimizing interference.
D. Multiple frequencies can be received simultaneously if desired.

E4C11. Why does input attenuation reduce receiver overload on the lower frequency HF bands with little or no impact on signal-to-noise ratio?
A. The attenuator has a low-pass filter to increase the strength of lower frequency signals.
B. The attenuator has a noise filter to suppress interference.
C. Signals are attenuated separately from the noise.
D. Atmospheric noise is generally greater than internally generated noise even after attenuation.

E4C12. How does a narrow-band roofing filter affect receiver performance?
A. It improves sensitivity by reducing front-end noise.
B. It improves intelligibility by using low Q circuitry to reduce ringing.
C. It improves blocking dynamic range by attenuating strong signals near the receive frequency.
D. All these choices are correct.

E4C13. What is reciprocal mixing?
A. Two out-of-band signals mixing to generate an in-band spurious signal.
B. In-phase signals cancelling in a mixer resulting in loss of receiver sensitivity.
C. Two digital signals combining from alternate time slots.
D. Local oscillator phase noise mixing with adjacent strong signals to create interference to desired signals.

E4C14. What is the purpose of the receiver IF Shift control?
A. To permit listening on a different frequency from the transmitting frequency.
B. To change frequency rapidly.
C. To reduce interference from stations transmitting on adjacent frequencies.
D. To tune in stations slightly off frequency without changing the transmit frequency.

Answer Key

E4B01: B	E4B02: A	E4B03: C	E4B04: A	E4B05: B
E4B06: D	E4B07: A	E4B08: C	E4B09: B	E4B10: B
E4B11: D	E4C01: D	E4C02: A	E4C03: C	E4C04: D
E4C05: B	E4C06: D	E4C07: B	E4C08: D	E4C09: C
E4C10: C	E4C11: D	E4C12: C	E4C13: D	E4C14: C